

12

Symmetry and Patterns

Tune In

Go to MyInspire for  

In India, block printing originated in Rajasthan, Gujarat, and Madhya Pradesh. The art form has become a major source of Indian fashion.



Look at the given picture, the woman has just started making a pattern on a cloth using the wooden blocks.



The wooden block has a design carved on it. The blocks are dipped in different colours and then stamped on the cloth to form a block printing.

In the space given below, extend the same pattern making by the woman in the above picture.

Blank space for extending the pattern.

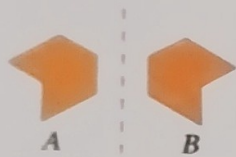
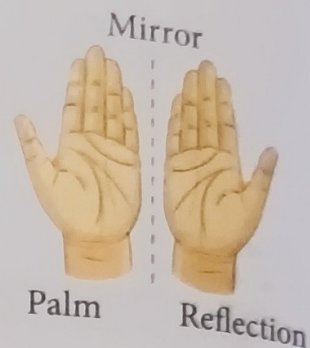
REFLECTION

When we look into a mirror, we see our reflection.

Have you read the story of the greedy dog which lost its bone when it barked at its reflection in the water? The reflection of an object is called its image.

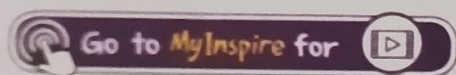
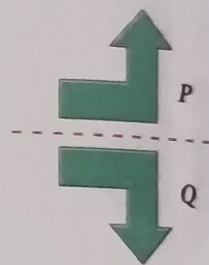
Hold your palm in front of a mirror. Look at the reflection in the mirror. What do you observe? The size of the image is the same as that of the object and it is at the same distance from the central line which acts like the edge of a mirror.

Here are some more pictures that show reflection.



Here, A, which is the object, is reflected along the vertical line. B is the mirror image of A or the reflection of A.

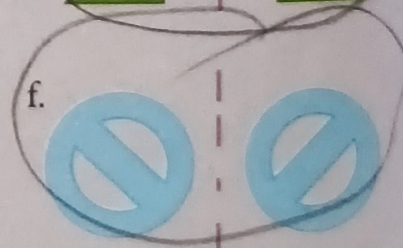
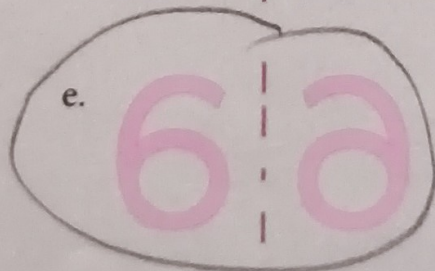
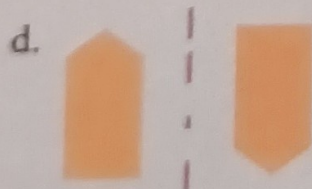
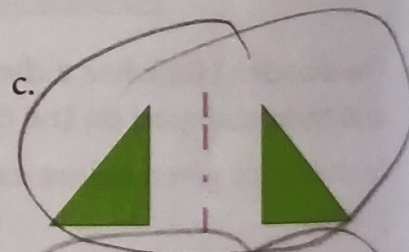
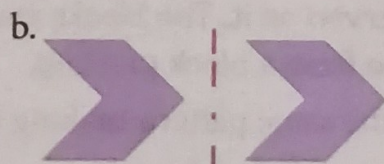
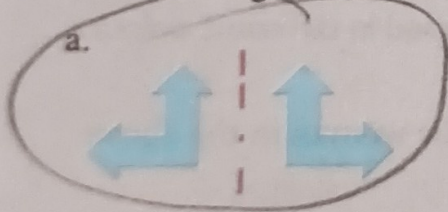
In the example here, the object shown as P is reflected across a horizontal line. Q is the mirror image of P.



Exercise 12A

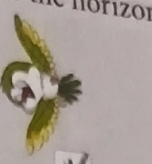
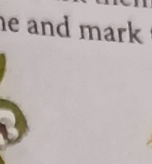
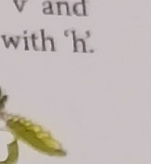
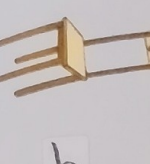
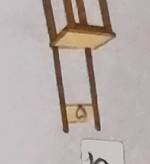
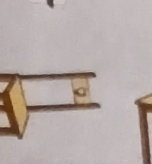
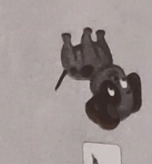
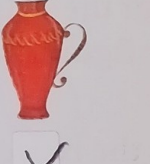
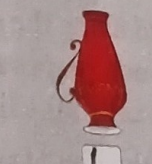
Go to MyInspire and attempt questions 1 and 2.

3. Circle the figures that are reflections of each other.



Application Based Questions

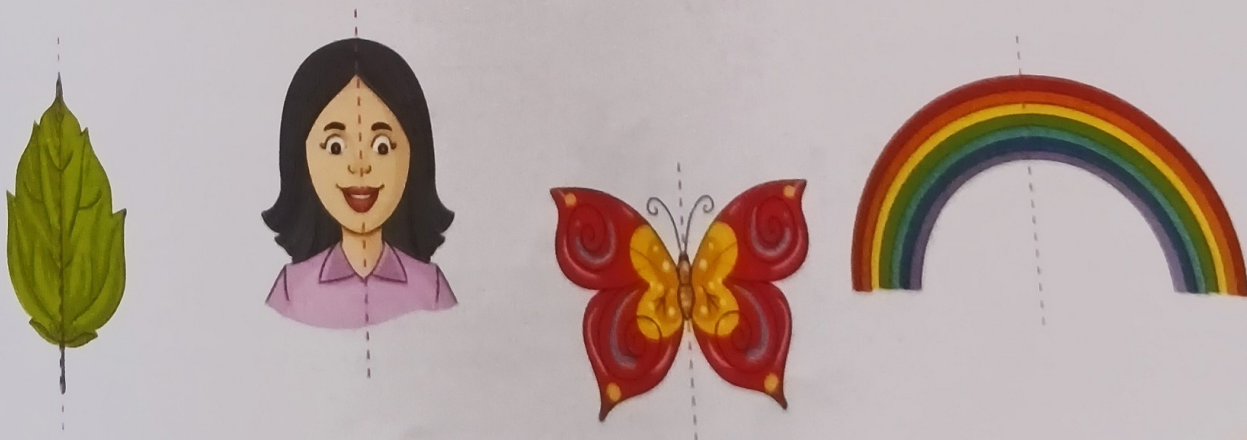
Choose the pair that are reflections across a vertical line and mark them with 'v' and choose the pair that are reflections across the horizontal line and mark them with 'h'.

- a.     
- b.     
- c.     
- d.     

SYMMETRY

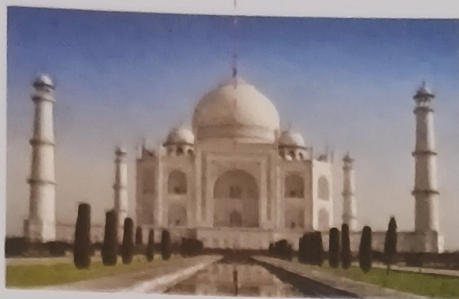
Symmetry in Nature

In everyday life, we see many objects which when divided from the centre, look exactly the same. It is called **symmetry**. Look at the following examples of symmetry.

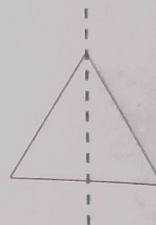
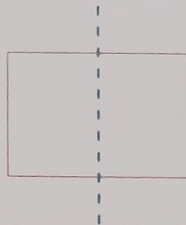
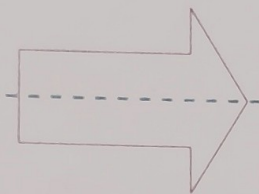
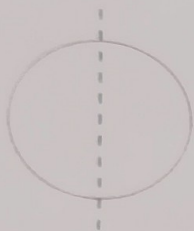


Man-Made Symmetry

A beautiful example of symmetry designed by human beings is the Taj Mahal. The arches found in many historical monuments are examples of perfect symmetry.



Now observe the following figures.

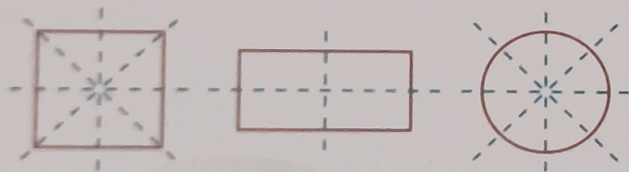


In the above figures, each shape is folded along the dotted line, one half of the shape falls on the other half and covers it exactly. Such shapes are called **symmetric figures**, and the line along which they could be folded is called the **line of symmetry**.

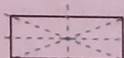
When we fold a symmetric shape along the line of symmetry, one side is the reflection of the other.

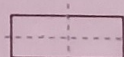
Some shapes have more than one line of symmetry.

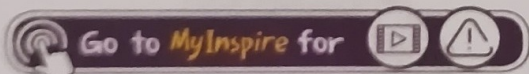
You can see from the given figures that a square has 4 lines of symmetry, while a rectangle has two. You can see that a circle has countless lines of symmetry because every diameter is a line of symmetry.



Watch Out!

 has 4 lines of symmetry. (✗)

 has 2 lines of symmetry. (✓)



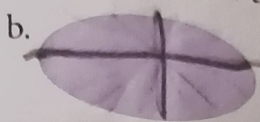
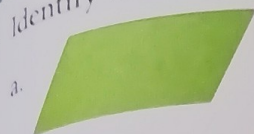
Teacher's Note

By placing the cut-out of a figure on the board on which the central line has been drawn, explain to the students that the central line is at the same distance from the object and the image. Show them how the image is obtained by flipping the cut-out across the central line.

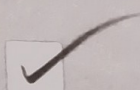
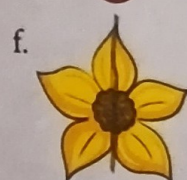
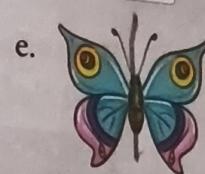
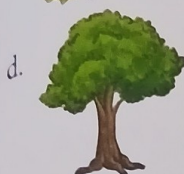
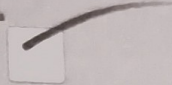
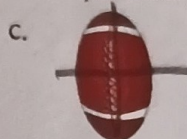
Exercise 12B

Go to MyInspire and attempt question 1.

Identify the symmetric shapes and draw the line or lines of symmetry.

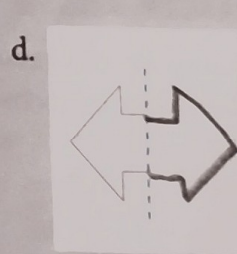
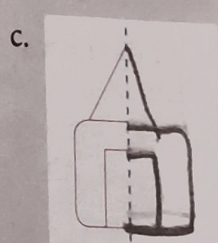
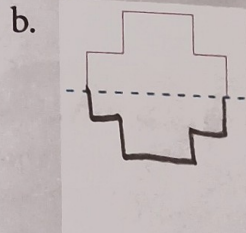
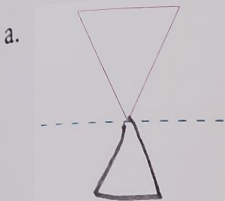


Tick (✓) the symmetrical figures. Also, draw the line or lines of symmetry.



Application Based Questions

Draw the reflection of each shape.



TESSELLATIONS

Tessellation is fitting geometrical shapes without any gaps or overlapping to form a repeating pattern. Tessellation is used in tiles for flooring and hence is also known as tiling. Tessellations can be found in nature, art and architecture, games, and so on. Let us see some examples.

1. In nature

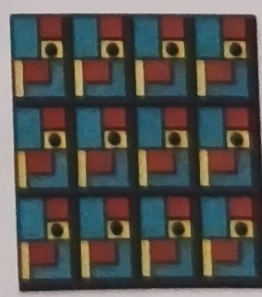


Honeycomb



Pineapple

2. In art and architecture



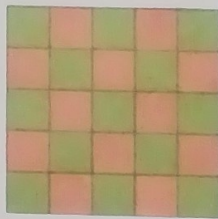
Floor

Tessellation patterns can be made in many ways—using the same polygon or with a combination of different polygons. There are also tessellations created with shapes other than polygons, such as circles and curves.

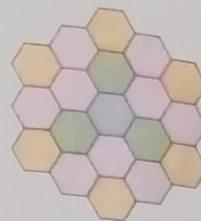
Tessellations with the Same Shape



Triangles

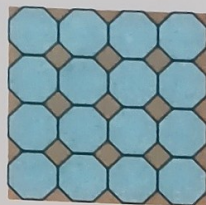
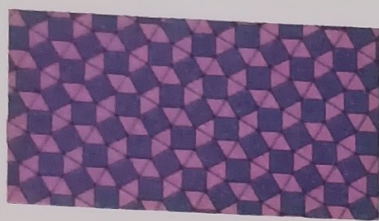


Squares

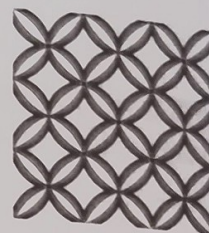
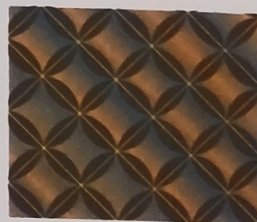


Hexagons

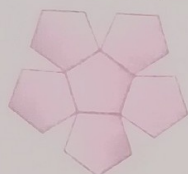
Tessellations with More than One Shape



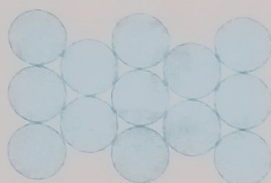
Tessellations with Circles, Curves, and Polygons



Some shapes do not tessellate on their own. A few examples are given below.



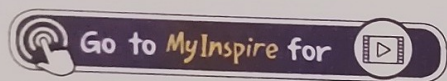
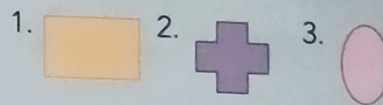
Pentagons



Circles

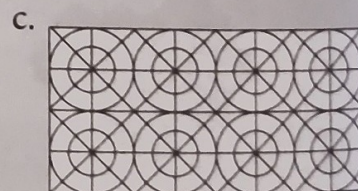
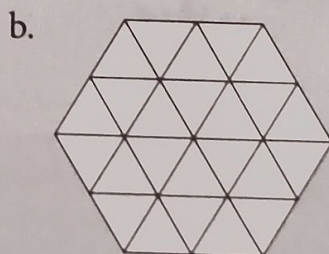
Let's Try! 1

Which of the following shapes do not tessellate?



Exercise 12C

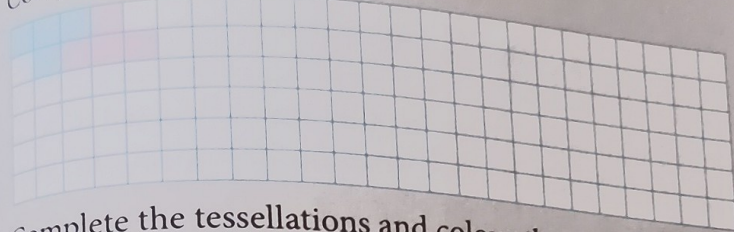
- Identify the following types of tessellations and complete the table. Colour each tessellation using colours of your choice. Also name the shapes used in tessellations.



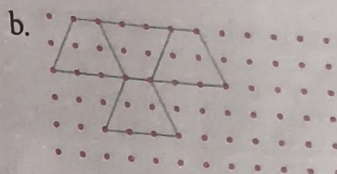
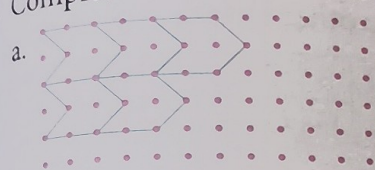
Type of Tessellation	Name of the shapes used
a.	
b.	
c.	

Application Based Questions

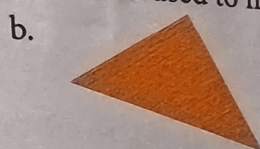
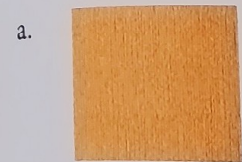
2. Complete the following tessellation.



3. Complete the tessellations and colour them.



4. Take a square chart paper of side 30 cm. Without actual measurement can you guess which tile among the three could be used to fill the square?



Now trace the tile you have chosen on a thick sheet of paper, cut it out and create a tessellation pattern on the chart paper.

Were you able to cover the complete chart paper?

PATTERNS

Alphabetic Patterns

Patterns can be created using the English alphabet.

Let us create some patterns using the letters of the English alphabet.

A B B A B B A B B A....

A B A A B A A A B A A A B ...

X Y Z X Y Z X Y Z X Y Z ...

Information can occasionally be kept private. Therefore, rather than putting it in plain language, it is coded by using numbers or alternate letters in place of the original characters. So you can code and decode secret messages by understanding the pattern they follow. Follow the pattern in which each letter is given a number.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26

You can use this code to write different coded messages.

Let us see some examples.

8 5 12 12 15 means HELLO.

3 15 13 5 19 15 15 14 means COME SOON.

In other codes, letters are substituted with other letters. In other words, instead of A, write Z; instead of B, write Y—that is, instead of the first letter, write the 26th letter, and instead of the second letter, write the 25th letter.

Follow the pattern in which each letter is substituted with the other letter.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A

Let us see some examples.

HAVE A GOOD DAY is written as SZE V Z TLLW WZB.

FANTASTIC is written as UZMGZHGRX.

Number Patterns

We can find the patterns in multiplication tables also.

Let us see some examples.

$5 \times 1 = 5$	$9 \times 1 = 9$
$5 \times 2 = 10$	$9 \times 2 = 18$
$5 \times 3 = 15$	$9 \times 3 = 27$
$5 \times 4 = 20$	$9 \times 4 = 36$
$5 \times 5 = 25$	$9 \times 5 = 45$
$5 \times 6 = 30$	$9 \times 6 = 54$
$5 \times 7 = 35$	$9 \times 7 = 63$
$5 \times 8 = 40$	$9 \times 8 = 72$
$5 \times 9 = \underline{\hspace{2cm}}$	$9 \times 9 = 81$
$5 \times 10 = \underline{\hspace{2cm}}$	$9 \times 10 = 90$
$5 \times 11 = \underline{\hspace{2cm}}$	$9 \times 11 = 99$
$5 \times 12 = \underline{\hspace{2cm}}$	$9 \times 12 = 108$
$5 \times 13 = \underline{\hspace{2cm}}$	$9 \times 13 = 117$
$5 \times 14 = \underline{\hspace{2cm}}$	$9 \times 14 = \underline{\hspace{2cm}}$

In the table of 5, observe the ones and tens place. In the ones place, we have 5, 0, and so on. Whereas in the tens place, we have 1 and 1, then 2 and 2, 3 and 3, 4 and 4, and so on.

In the table of 9, observe the ones and tens place. In the ones place, the digits go on decreasing till 0 and then again from 9 to 0. Whereas in the tens place, the digits go on increasing till 9. Thus, the next numbers again increase from 9 to 10 and 10 to 11, and so on.

plain
characters.
follow.

X	Y	Z
24	25	26

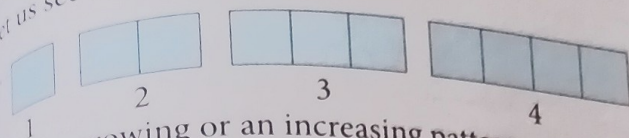
of A, write
and instead

X	Y	Z
C	B	A

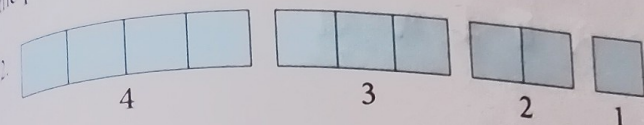
nes and
e have 5,
ns place,
3 and 3, 4

nes and
e digits
n again
s place,
9. Thus,
se from 9

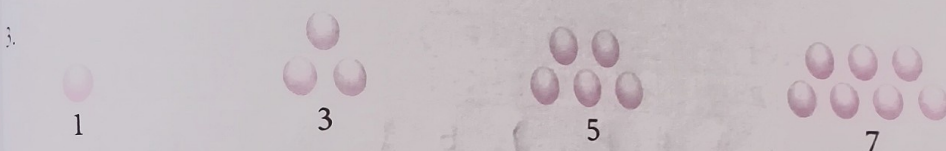
Picture patterns can also be connected to number patterns.
Let us see some examples.



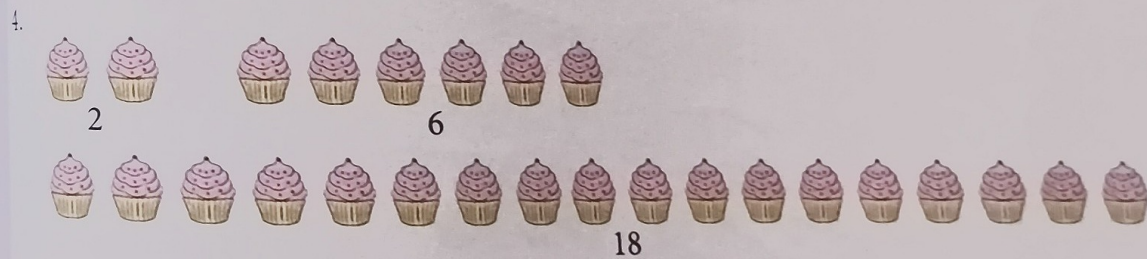
This is a growing or an increasing pattern.
The number pattern is 1, 2, 3, 4.
The rule here is to add 1 to get the next number in the pattern.



This is a decreasing pattern.
The number pattern is 4, 3, 2, 1
The rule here is to subtract 1 to get the next number.



This is an increasing pattern.
The number pattern is 1, 3, 5, 7
Here the rule is to add 2 to get the next number.



This is an increasing pattern.
The number pattern is 2, 6, 18.
Here the rule is to multiply by 3 to get the next number.

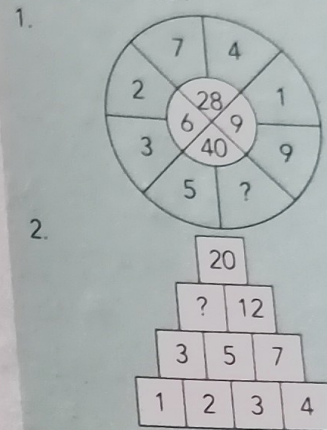
Go to **MyInspire** for

Teacher's Note

The students should be encouraged to appreciate the patterns and tessellations they come across in their daily life.

Let's Try! 2

Observe the pattern and find the missing number.



Exercise 12D

Go to MyInspire and attempt question 1.

2. Use the below code to decode the given messages.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26

a. 19 5 5 25 15 21 20 15 13 15 18 18 15 23
S E E Y O U T O M O R R O W

b. 7 15 15 4 13 15 18 14 9 14 7
G O O D M O R N I N G

3. Write these using the first code.

a. GOOD JOB 7 15 15 4 10 15 2

b. SAVE WATER 19 1 22 5 2 1 20 5 18

c. GET WELL SOON 7 5 20 23 5 12 12 9 15 15 14

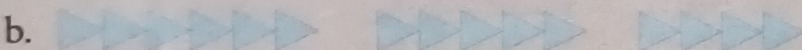
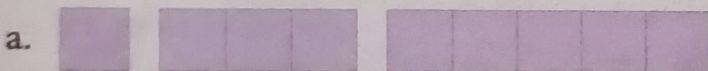
4. Use the below code to decode the given messages.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A

a. SZKKB YRIGSWZB H A P P Y B I R T H D A Y

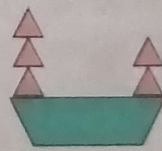
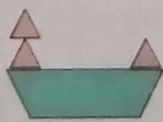
b. NVVG BLF HLLM M E E T Y O U S O O N

5. What will come next in the pattern? Draw the next two shapes. Write the number pattern to match the pattern and write the rule.



Application Based Questions

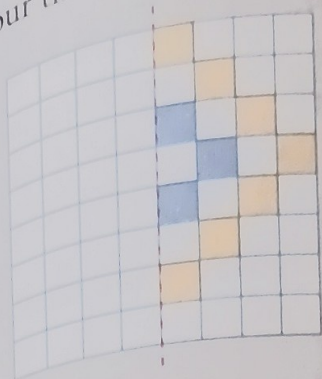
6. Find and correct the errors in the following patterns.



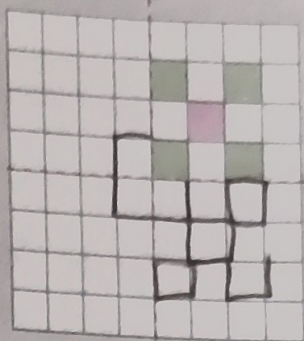
b. 135, 245, 355, 465, 555, 685, 795, 905

c. 10000, 9950, 9900, 9800, 9750, 9700

Colour the squares to complete the following symmetrical patterns.



b.



Teachers are the role models of society as they help develop our knowledge and talents. Nikhil showed appreciation to his class and subject teachers by giving them a 'Thank You card'. He wrote a message on the card and design that using the pattern of the flowers as shown below. Using the same flowers drawn by Nikhil-yellow and pink colour, create as many different patterns as possible.

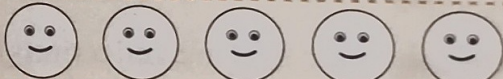


Let's Review

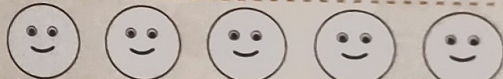


Come here after you have completed the chapter. Colour 1 to 5 😊 to show how well you understand a topic. (5 😊 = Very well; 1 😊 = Not at all)

Reflection



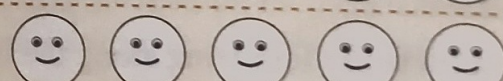
Symmetry



Tessellations



Patterns



Revision Exercises

Go to MyInspire for



Start Up



1. Observe the pattern and write the next two steps.

$$0 \times 9 + 8 = 8$$

$$9 \times 9 + 7 = 88$$

$$98 \times 9 + 6 = 888$$

$$987 \times 9 + 5 = 8888$$

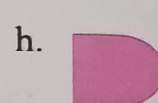
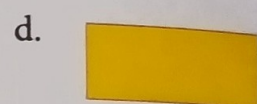
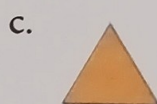
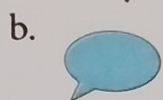
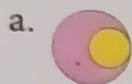
2. Observe the pattern and write the next two steps.

$$1 + 3 = 4 = 2 \times 2$$

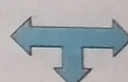
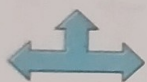
$$1 + 3 + 5 = 9 = 3 \times 3$$

$$1 + 3 + 5 + 7 = 16 = 4 \times 4$$

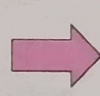
3. Draw the lines of symmetry of the following figures, if any.



4. Circle the pair that are reflections across a vertical line.



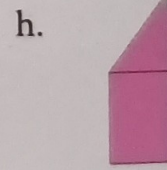
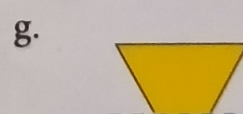
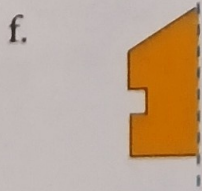
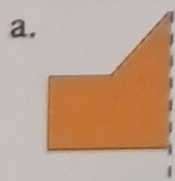
5. Circle the pair that are reflections across a horizontal line.



Level Up



1. Complete the following along the line of symmetry.



Look at the reflection given below. Find out the sentence.

ALL WORK AND NO PLAY MAKES JACK A DULL BOY

Create a slogan on pollution control and write its mirror reflection.
Use the below code to decode the given messages.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26

a. 19 5 5 25 15 21

20 15 13 15 18 18 15 23

b. 9 12 15 22 5

13 25 9 14 4 9 1

5. Write these using the first code.

a. MATHEMATICS

b. SAVE ENERGY

HOTS Questions



1. How many lines of symmetry does a circle have?



2. Draw the lines of symmetry for the given figure.



Everyday Maths



We observe symmetry and tessellations in many places in our daily lives, however we do not realize it at first sight. There are plenty of examples of symmetry in arts, buildings, and monuments. Whereas tessellations can be noticed on tiled surfaces, honeycomb etc. In nature, both—symmetry and tessellations, contributes to the beauty of things.

1. Identify the various objects you come across in your daily life which have line or lines of symmetry. Collect the pictures of such objects and paste them in your notebook.
2. Identify the various items you come across in your daily life which are tessellations. Collect the pictures of such objects and paste them in your notebook.